

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented multi-agent workflows using **Python** and **LangGraph** for an AI-powered code generation assistant, orchestrating specialist agents for code analysis and synthesis.
- Developed **context engineering** techniques, including dynamic system prompt generation and agent skill customization, to improve LLM reasoning accuracy by approximately **18%** on benchmark tasks.
- Built and evaluated experimental AI outputs, focusing on **prompt engineering evaluation** and red-teaming prompts to identify failure modes in LLM-generated code suggestions.
- Contributed to an internal **AI tooling landscape** by developing prototypes for new agentic patterns, enabling rapid experimentation and feedback cycles within the engineering team.
- Applied software engineering best practices, including **TDD** and **CI/CD**, to the development of AI agent components, ensuring code quality and maintainability.
- Integrated LLM reasoning capabilities into existing Java Spring Boot services, focusing on managing **context window limitations** and optimizing prompt response times.
- Containerized AI workflow components using **Docker** and deployed them to a Kubernetes cluster, ensuring scalable and reliable execution of agent-based systems.

SKILLS

AI/ML – Multi-agent workflows, Context engineering, Prompt engineering evaluation, LLM reasoning, Agent customization, System prompts, Agent skills, MCP, Red-teaming prompts

Languages – Python, Java, Node.js, Go

Frameworks/Libraries – LangGraph, CrewAI, AutoGen, scikit-learn, Spring Boot

Development Practices – TDD, CI/CD, Code reviews, Agile methodologies

Cloud & Containerization – Docker, Kubernetes, AWS, Linux

PROJECTS

Agentic Workflow for Code Refactoring

- Developed a multi-agent system using **Python** and **AutoGen** to automate code refactoring tasks, orchestrating agents for static analysis, suggestion generation, and automated testing.
- Implemented context management strategies to feed relevant code snippets and historical data to LLMs, improving the quality and relevance of refactoring suggestions by **~22%**.
- Designed agent skills and system prompts to guide the LLM in understanding and applying specific refactoring patterns, reducing manual intervention by an estimated **35%**.

LLM-Powered Developer Productivity Tool

- Built a prototype application using **FastAPI** and **LangChain** to provide developers with AI-assisted code completion and documentation generation.
- Integrated with a large language model API to process developer queries and generate contextually relevant code snippets and explanations, achieving a user satisfaction rating of **8.2/10** in internal testing.
- Focused on prompt optimization and retrieval-augmented generation (RAG) techniques to enhance the accuracy and usefulness of AI-generated content for common coding tasks.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)